Week 16 Progress Report

Details

- Name of Student: Tirthak Girish Likhar
- PRN: 22070521041
- Name of Industry: SS Infotech Pvt. Ltd.
- **Reporting Week: (20/04/2026 to 24/04/2026)**
- Institute Mentor Name: Dr. Shreyas Hole
- Industry Mentor Name: Mr. Viraj Patle

Overview of the Week

- Week 16 focused on running a Security Configuration Assessment (SCA) compliance scan on the Windows 11 monitored endpoint (LAPTOP-MS9PL5OA, Agent 002) using the Wazuh SCA module. SCA is a built-in Wazuh feature that automatically audits endpoint configurations against industry-standard security benchmarks and reports compliance posture in real time.

- The scan was executed automatically by the Wazuh agent against the CIS Microsoft Windows 11 Enterprise Benchmark v1.0.0 policy a widely recognised hardening standard published by the Center for Internet Security (CIS). The Wazuh Dashboard provided a full breakdown of 395 checks covering password policies, account lockout settings, audit policies, user rights, and system configurations.

- The Windows 11 endpoint scored 32% compliance with 127 checks passed, 259 failed, and 9 not applicable. The majority of failed checks were in the areas of password policy hardening, account lockout configuration, and Windows security settings that are not enforced by default on a standalone home edition machine. The results were reviewed, documented, and used to identify key remediation areas for the SOC lab.

Objectives of the Week

- To run the Wazuh SCA module on the Windows 11 endpoint (LAPTOP-MS9PL5OA) and generate a compliance report against the CIS Microsoft Windows 11 Enterprise Benchmark v1.0.0 policy.

- To review the SCA scan results on the Wazuh Dashboard — overall score, total checks, passed, failed, and not applicable counts.

- To identify and document key failed checks representing security misconfigurations on the Windows endpoint.

- To identify passed checks demonstrating baseline security controls already in place.

- To analyse the compliance gap and understand remediation steps that would be required to improve the score in a real SOC environment.

- To resolve the Wazuh agent disconnection issue caused by UFW blocking ports 1514 and 1515, restoring full agent-manager communication before running the SCA scan.

Key Accomplishments

- Resolved Wazuh agent disconnection by opening UFW ports 1514 (tcp/udp) and 1515 (tcp) on the Ubuntu Server — agent LAPTOP-MS9PL5OA (002) returned to Active status.

- Successfully executed SCA scan via the Wazuh Dashboard
  Agents → LAPTOP-MS9PL5OA → SCA → CIS Microsoft Windows 11 Enterprise Benchmark v1.0.0.

- SCA scan completed with 395 total checks evaluated against the CIS Windows 11 Enterprise Benchmark — score: 32% (127 passed, 259 failed, 9 not applicable).

- Identified key failed checks in password policy (enforce password history, minimum length, complexity), account lockout threshold, and registry-based security settings.

- Identified passed checks confirming baseline controls: maximum password age, account lockout duration, lockout counter reset, and Administrator account disabled status.

- Reviewed and documented SCA findings as a compliance posture baseline for the Windows 11 endpoint, mapping the gap between current configuration and CIS hardening requirements.

Issues Faced & Resolutions – Week 16

1. Wazuh agent (LAPTOP-MS9PL5OA) showing Disconnected on Dashboard startup

- **Issue:** After booting the lab, Agent 002 (LAPTOP-MS9PL5OA) appeared as Disconnected on the Wazuh Dashboard even though the Windows Wazuh service was running. The agent_control -l output on the Ubuntu Server also confirmed Disconnected status.

- **Resolution:** Identified that UFW on the Ubuntu Server was blocking ports 1514 and 1515 the Wazuh agent communication ports due to rules added during Week 15 troubleshooting. Ran: sudo ufw allow 1514/tcp && sudo ufw allow 1514/udp && sudo ufw allow 1515/tcp && sudo ufw reload. Then restarted the Wazuh Manager and Windows agent service. Agent 002 returned to Active status within 30 seconds.

Tools Used – Week 16

Tool / Technology	Purpose
Wazuh SCA Module v4.7.5	Automated security configuration assessment against CIS benchmark policy
Wazuh Dashboard v4.7.5	Viewing SCA scan results, scores, and individual check details
Wazuh Manager v4.7.5	Running the SCA scan and collecting results from the Windows agent
CIS Microsoft Windows 11 Enterprise Benchmark v1.0.0	Compliance policy used as the assessment baseline
Windows 11 Home (LAPTOP-MS9PL5OA)	Monitored endpoint — Wazuh Agent 4.7.5, Agent ID 002
VirtualBox 7.x	Hypervisor hosting the Ubuntu Server VM
Ubuntu Server 24.04	Wazuh Server — Manager, Indexer, and Dashboard (192.168.1.6)

Conclusion of Week 16

Week 16 successfully demonstrated the use of Wazuh's built-in Security Configuration Assessment module to evaluate the compliance posture of the Windows 11 monitored endpoint against the CIS Microsoft Windows 11 Enterprise Benchmark v1.0.0.

The scan revealed a 32% compliance score a realistic result for a standalone Windows 11 Home machine that has not been hardened beyond default settings. In a real enterprise SOC, SCA results like these would be escalated to the IT or security team as a baseline report, with failed checks prioritised for remediation based on risk severity.

The exercise demonstrated how automated compliance scanning provides SOC analysts with actionable visibility into endpoint security posture without requiring manual audits. Combined with the threat detection and active response capabilities built in earlier weeks, the SOC lab now covers the full spectrum from proactive compliance monitoring to reactive incident response.

SCA Scan Summary

Parameter	Value
Policy	CIS Microsoft Windows 11 Enterprise Benchmark v1.0.0
Agent	LAPTOP-MS9PL5OA (Agent ID 002)
Operating System	**Microsoft Windows 11 Home Single Language 10.0.26200.8037**
Score	32%
Total Checks	395
Passed	127
Failed	259
Not Applicable	9
Last Scan	Mar 25, 2026 @ 03:06:01.000

Notable Failed Checks

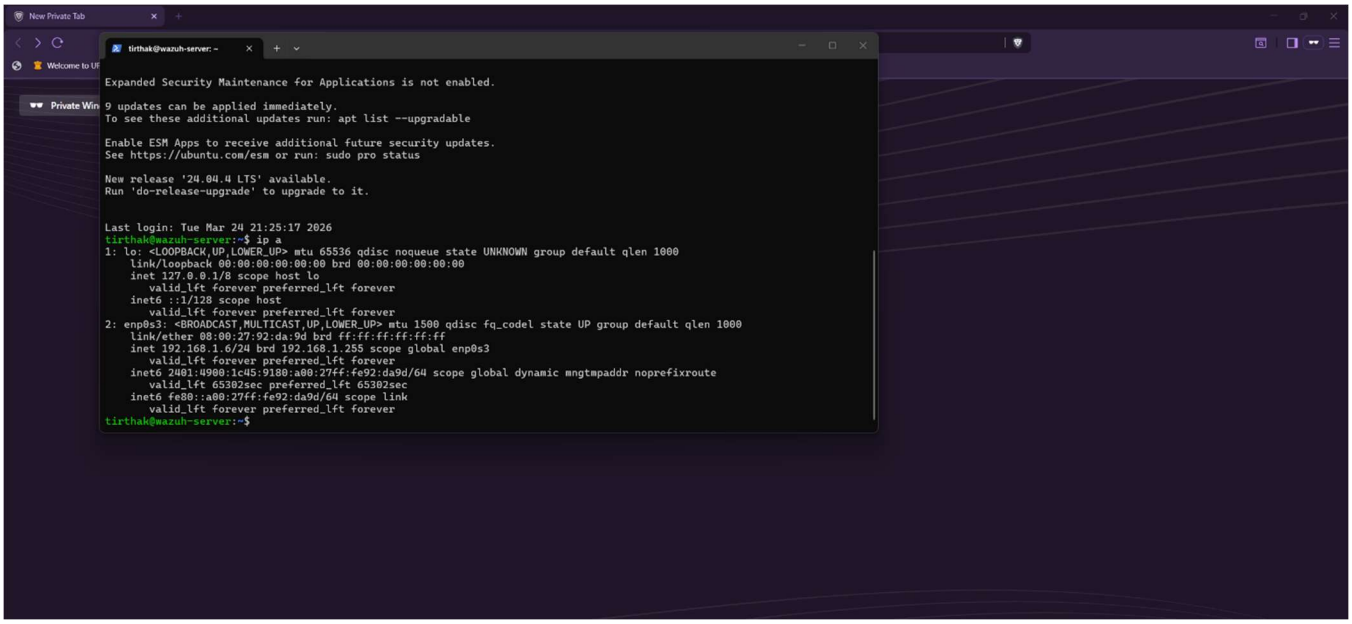
Check ID	Title	Target
26000	Ensure 'Enforce password history' is set to '24 or more password(s)'	Command: net.exe accounts
26002	Ensure 'Minimum password age' is set to '1 or more day(s)'	Command: net.exe accounts
26003	Ensure 'Minimum password length' is set to '14 or more character(s)'	Command: net.exe accounts
26004	Ensure 'Password must meet complexity requirements' is set to 'Enabled'	Command: powershell Get-ADDefaultDomainPasswordPolicy
26005	Ensure 'Relax minimum password length limits' is set to 'Enabled'	Registry: HKLM\System\CurrentControlSet\Control\SAM
26007	Ensure 'Account lockout threshold' is set to '5 or fewer invalid logon attempt(s), but not 0'	Command: net.exe accounts

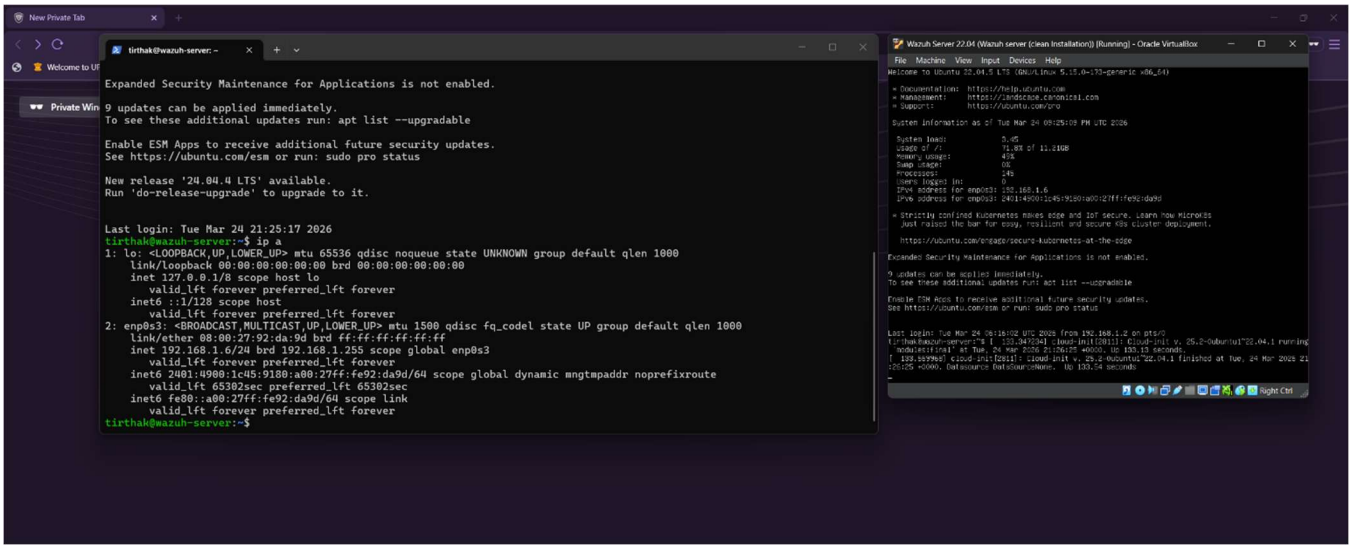
Notable Passed Checks

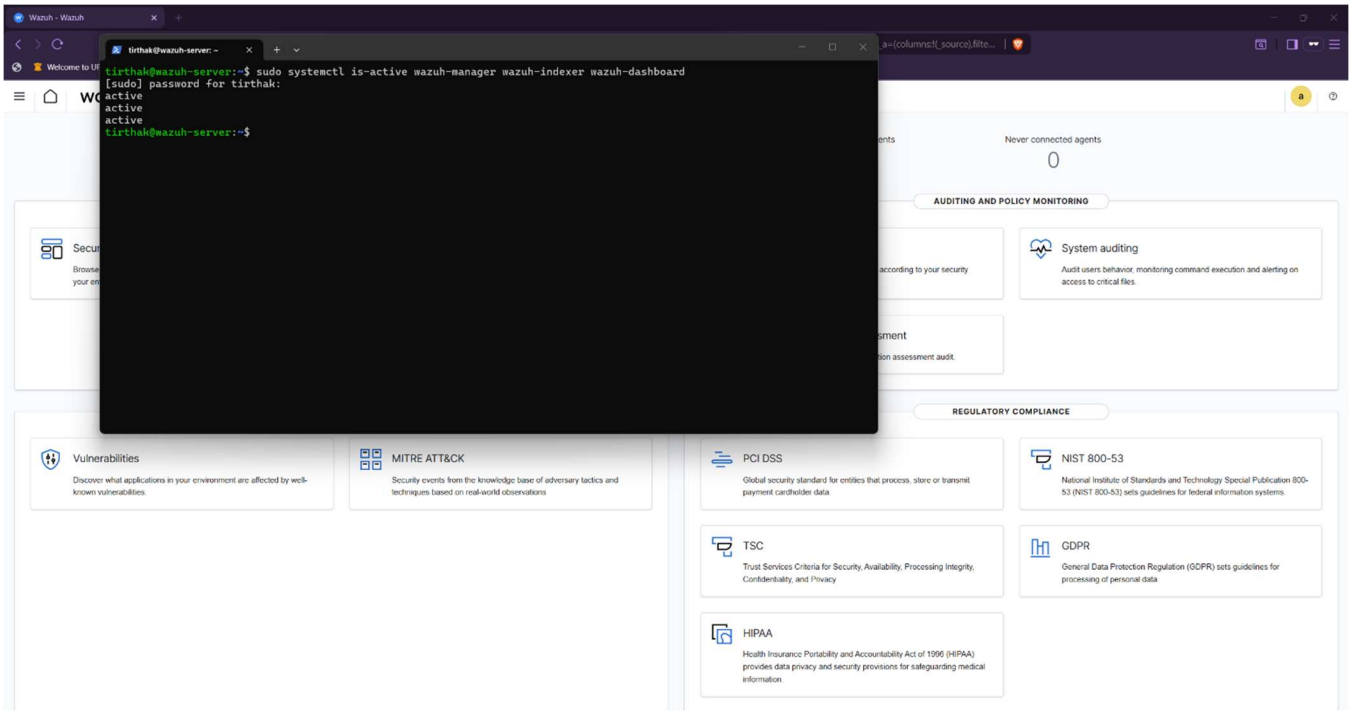
Check ID	Title	Target
26001	Ensure 'Maximum password age' is set to '365 or fewer days, but not 0'	Command: net.exe accounts
26006	Ensure 'Account lockout duration' is set to '15 or more minutes'	Command: net.exe accounts
26008	Ensure 'Reset account lockout counter after' is set to '15 or more minutes'	Command: net.exe accounts
26009	Ensure 'Accounts: Administrator account status' is set to 'Disabled'	Command: net user administrator

Screenshots of Activities (if Any)

[Wazuh Dashboard SCA tab showing CIS policy and 32% score, full check list with Failed/Passed results, UFW status showing ports 1514/1515 allowed, agent_control -l showing Agent 002 Active]








```
tirthak@wazuh-server:~$ sudo /var/ossec/bin/agent_control -l

Wazuh agent_control. List of available agents:
   ID: 000, Name: wazuh-server (server), IP: 127.0.0.1, Active/Local
   ID: 002, Name: LAPTOP-MS9PL5OA, IP: any, Disconnected

List of agentless devices:

tirthak@wazuh-server:~$
```

```
PS C:\Users\Tirthak> Restart-Service wazuhsvc
PS C:\Users\Tirthak> Start-Sleep -Seconds 5
PS C:\Users\Tirthak> Get-Service wazuhsvc

Status   Name               DisplayName
------   ----               -----------
Running  wazuhsvc           Wazuh


PS C:\Users\Tirthak>
```

```
tirthak@wazuh-server:~$ sudo /var/ossec/bin/agent_control -l

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   ID: 000, Name: wazuh-server (server), IP: 127.0.0.1, Active/Local
   ID: 002, Name: LAPTOP-MS9PL5OA, IP: any, Disconnected

List of agentless devices:

tirthak@wazuh-server:~$ sudo /var/ossec/bin/agent_control -l

Wazuh agent_control. List of available agents:
   ID: 000, Name: wazuh-server (server), IP: 127.0.0.1, Active/Local
   ID: 002, Name: LAPTOP-MS9PL5OA, IP: any, Disconnected

List of agentless devices:

tirthak@wazuh-server:~$
```
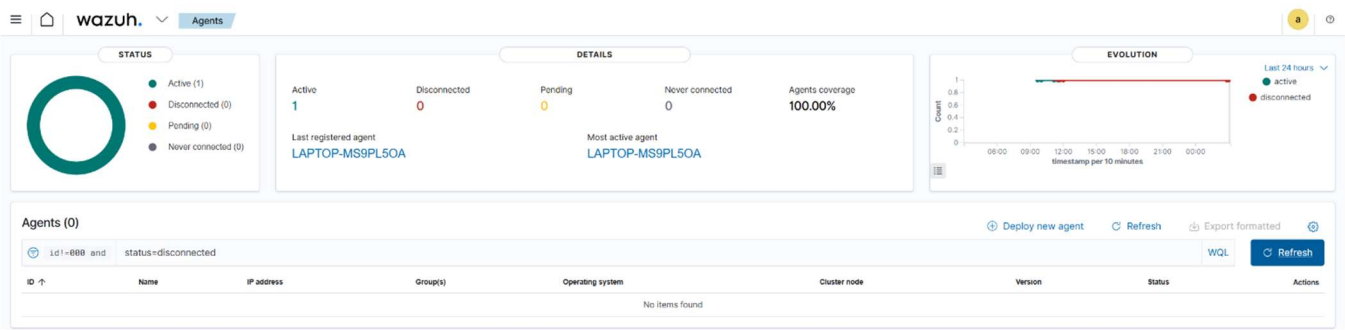
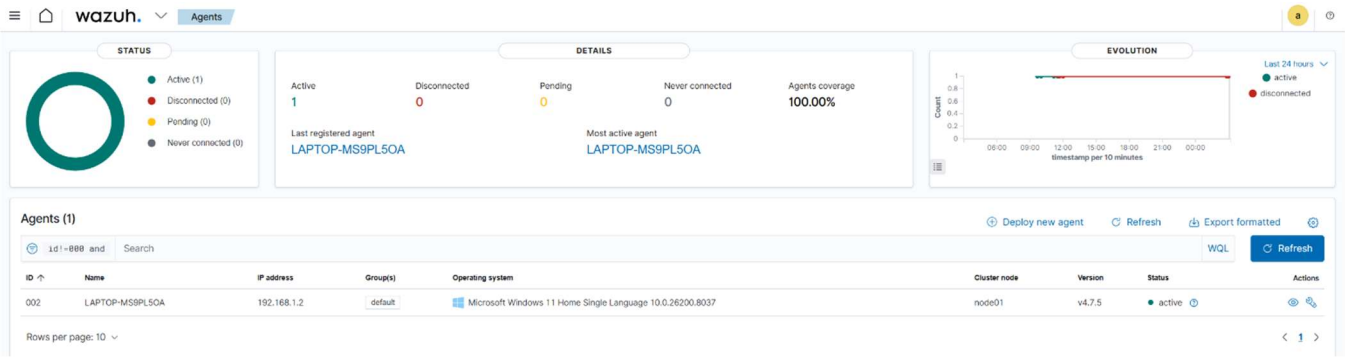

```
PS C:\Users\Tirthak> Stop-Service wazuhsvc
PS C:\Users\Tirthak> Start-Service wazuhsvc
PS C:\Users\Tirthak> Get-Service wazuhsvc

Status   Name               DisplayName
------   ----               -----------
Running  wazuhsvc           Wazuh


PS C:\Users\Tirthak>
```

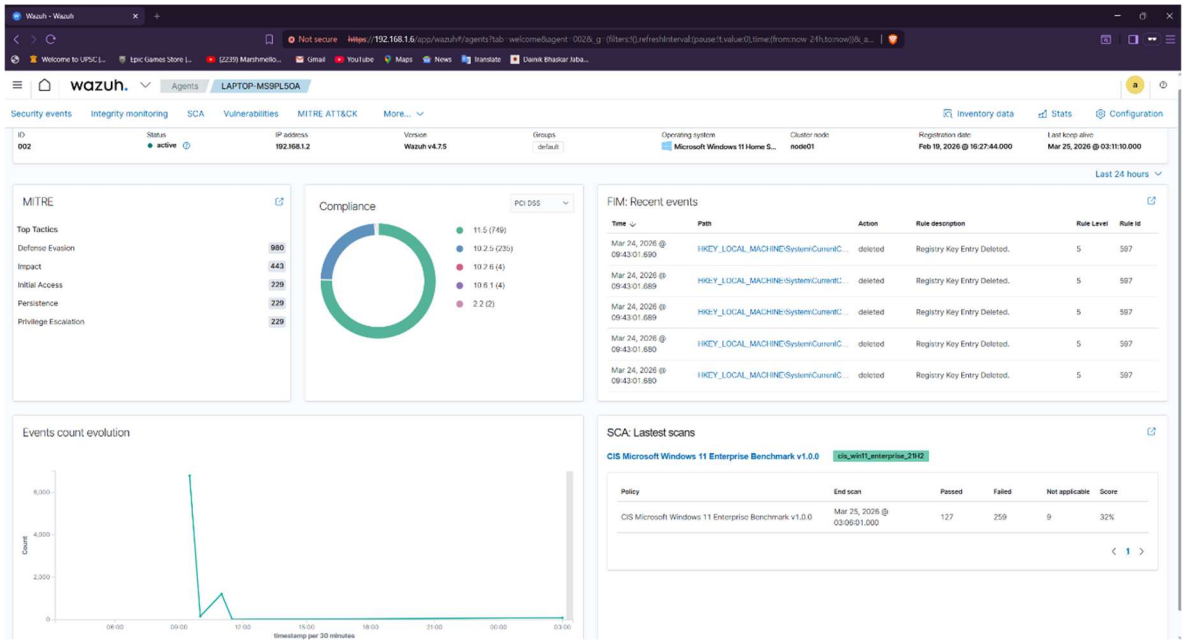
```
tirthak@wazuh-server:~$ sudo /var/ossec/bin/agent_control -l

Wazuh agent_control. List of available agents:
   ID: 000, Name: wazuh-server (server), IP: 127.0.0.1, Active/Local
   ID: 002, Name: LAPTOP-MS9PL5OA, IP: any, Disconnected

List of agentless devices:

tirthak@wazuh-server:~$
```

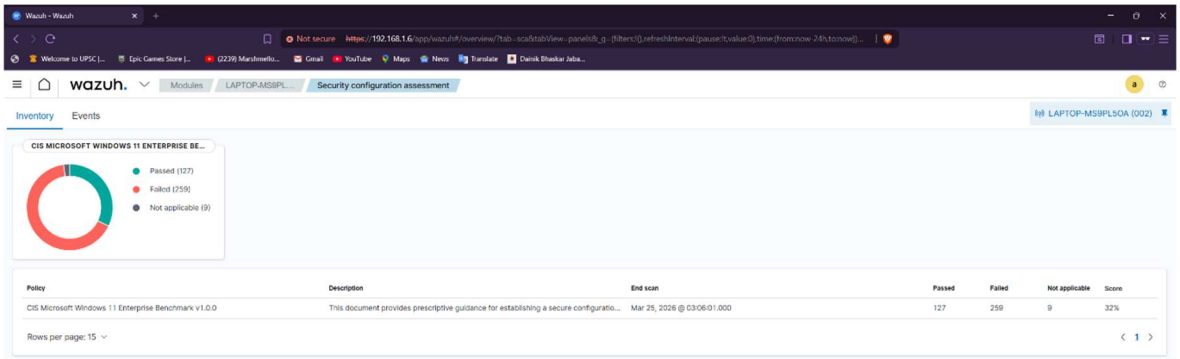
```
tirthak@wazuh-server: ~
List of agentless devices:

tirthak@wazuh-server:~$ sudo ufw allow 1514/tcp
sudo ufw allow 1514/udp
sudo ufw allow 1515/tcp
sudo ufw reload
sudo ufw status
Rule added
Rule added (v6)
Rule added
Rule added (v6)
Rule added
Rule added (v6)
Firewall reloaded
Status: active

To                         Action      From
--                         ------      ----
22/tcp                     ALLOW       Anywhere
443/tcp                    ALLOW       Anywhere
1514/tcp                   ALLOW       Anywhere
1514/udp                   ALLOW       Anywhere
1515/tcp                   ALLOW       Anywhere
22/tcp (v6)                ALLOW       Anywhere (v6)
443/tcp (v6)               ALLOW       Anywhere (v6)
1514/tcp (v6)              ALLOW       Anywhere (v6)
1514/udp (v6)              ALLOW       Anywhere (v6)
1515/tcp (v6)              ALLOW       Anywhere (v6)

tirthak@wazuh-server:~$
```

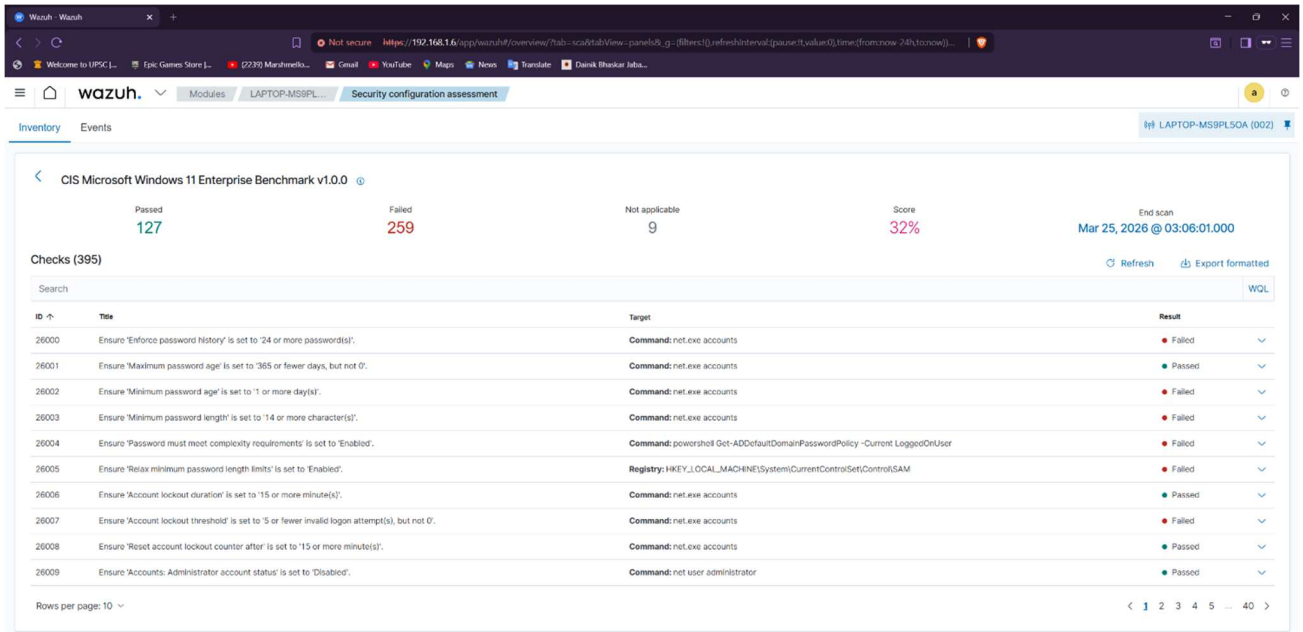
```
tirthak@wazuh-server: ~

Rule added (v6)
Rule added
Rule added (v6)
Firewall reloaded
Status: active

To                         Action      From
--                         ------      ----
22/tcp                     ALLOW       Anywhere
443/tcp                    ALLOW       Anywhere
1514/tcp                   ALLOW       Anywhere
1514/udp                   ALLOW       Anywhere
1515/tcp                   ALLOW       Anywhere
22/tcp (v6)                ALLOW       Anywhere (v6)
443/tcp (v6)               ALLOW       Anywhere (v6)
1514/tcp (v6)              ALLOW       Anywhere (v6)
1514/udp (v6)              ALLOW       Anywhere (v6)
1515/tcp (v6)              ALLOW       Anywhere (v6)

tirthak@wazuh-server:~$ sudo systemctl restart wazuh-manager
sleep 20
sudo /var/ossec/bin/agent_control -l

Wazuh agent_control. List of available agents:
   ID: 000, Name: wazuh-server (server), IP: 127.0.0.1, Active/Local
   ID: 002, Name: LAPTOP-MS9PL5OA, IP: any, Active

List of agentless devices:

tirthak@wazuh-server:~$
```






wazuh.
Agents
LAPTOP-MS9PL5OA
Security events
Integrity monitoring
SCA
Vulnerabilities
MITRE ATT&CK
More...
Inventory data
Stats
Configuration
MITRE
Top Tactics
Defense Evasion 980
Impact 443
Initial Access 229
Persistence 229
Privilege Escalation 229
Compliance
PCI DSS
FIM: Recent events
Last 24 hours
Events count evolution
SCA: Lastest scans
CIS Microsoft Windows 11 Enterprise Benchmark v1.0.0


wazuh.
Modules
LAPTOP-MS9PL...
Security configuration assessment
Inventory
Events
LAPTOP-MS9PL5OA (002)
CIS MICROSOFT WINDOWS 11 ENTERPRISE BE...
Passed (127)
Failed (259)
Not applicable (9)
Rows per page: 15


wazuh.
Modules
LAPTOP-MS9PL...
Security configuration assessment
Inventory
Events
LAPTOP-MS9PL5OA (002)
CIS Microsoft Windows 11 Enterprise Benchmark v1.0.0
Passed 127
Failed 259
Not applicable 9
Score 32%
End scan Mar 25, 2026 @ 03:06:01.000
Checks (395)
Refresh
Export formatted
Search
WQL
Rows per page: 10